# Qwen2.5-VL-3B-Instruct — end-to-end architecture

3.75 B parameters = vision encoder 632.0 M + patch merger 36.7 M + language model 3.09 B  (token embedding shared with the LM head)

Shapes  [rows, features]:  P = image patches of 14×14 px,  N = P / 4 visual tokens,
L = prompt length in tokens (text + N),  d = 2048 (LLM width),  V = 151,936 (vocab rows)
Example used in all figures: 448×336 px image + “What is in the image?”
→ P = 32×24 = 768,  N = 16×12 = 192,  L = 219

## VISION PATH — runs once per image

**Image**
any size H × W
RGB

**Image processor (CPU)**
resize: H, W → multiples of 28 px
(aspect ratio kept, pixel budget
min_pixels … max_pixels)
normalize (CLIP mean / std)
repeat the frame ×2 (temporal patch)
cut into 14×14 px patches

[P,1176]

**Patch embedding**
Conv3D 3→1280
kernel = stride
= (2, 14, 14)
≡ Linear 1176→1280
no bias

[P,1280]

**Vision Transformer ×32**
width 1280 · 16 heads × 80
2D RoPE on q, k (row, col)
window attention: 8×8 patches
full attention: blocks 7, 15, 23, 31
SwiGLU MLP 1280→3420→1280
bidirectional (no causal mask)

[P,1280]

**Patch merger (projector)**
2×2 neighbouring patches
→ 1 token
RMSNorm, concat 4×1280 = 5120
Linear 5120→5120 · GELU
Linear 5120→2048

[N, 2048]

## TEXT PATH

**Prompt**
question
+ image

**Chat template**
<|im_start|>system … <|im_end|>
<|im_start|>user
<|vision_start|><|image_pad|><|vision_end|>
What is in the image?<|im_end|>
<|im_start|>assistant
<|image_pad|> is repeated N times

text

**Tokenizer**
byte-level BPE
151,643 tokens
+ 22 special
= 151,665 ids

[L] ids

**Token embedding**
lookup table V × d
= 151,936 × 2048
rows ≥ 151,665: unused padding
(text tokens → vectors)

[L,2048]

**Insert image features**
the N rows at <|image_pad|>
positions are replaced
by the N merger outputs

[L,2048]

## LANGUAGE MODEL

**3D position ids (M-RoPE)**
text: t = h = w = position
image: t shared, h = row, w = col
from token ids + image grid
→ cos / sin used in every layer

[3, L]

**Decoder layer ×36**
pre-RMSNorm + residual
self-attention (GQA, causal)
16 query heads, 2 KV heads
head dim 128, M-RoPE on q, k
SwiGLU MLP 2048→11008→2048
KV cache for decoding

[L, 2048]

**Final RMSNorm**

[1, 2048]  last pos.

**LM head**
Linear 2048 → 151,936
no bias
weights tied to token embedding

[1, 151,936] logits

**Next token**
softmax → greedy / sampling
only the last position
is needed

*decode loop: the new token is embedded and fed back,*
*one token per step; earlier keys / values come from the KV cache*

### Sizes of this model (from config.json)

```
Vision encoder (same shape in 3B and 7B)
  width 1280, 32 blocks, 16 heads × 80
  MLP 1280→3420→1280, window 112 px
Language model
  width d = 2048, 36 layers
  16 q-heads / 2 KV-heads × 128
  MLP 2048→11008→2048
  vocab rows 151,936, tied head: yes
  RoPE θ = 1,000,000, RMSNorm ε = 1e-06
```

### Where the parameters are



- ViT (patch embed + 32 blocks): 632.0 M  (16.8 %)
- patch merger: 36.7 M  (1.0 %)
- token embedding = LM head: 311.2 M  (8.3 %)
- 36 decoder layers + final norm: 2.77 B  (73.9 %)

bf16 weights: 7.5 GB  ·  KV cache: 36 KiB per token  ·  decoder ≈ 6.2 GFLOPs per generated token (linear layers)

Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# Qwen2.5-VL-7B-Instruct — end-to-end architecture

8.29 B parameters = vision encoder 632.0 M + patch merger 44.6 M + language model 7.62 B

Shapes [rows, features]: P = image patches of 14×14 px, N = P / 4 visual tokens,
L = prompt length in tokens (text + N), d = 3584 (LLM width), V = 152,064 (vocab rows)
Example used in all figures: 448×336 px image + "What is in the image?"
→ P = 32×24 = 768, N = 16×12 = 192, L = 219

## VISION PATH — runs once per image

**Image**
any size H × W
RGB

→

**Image processor (CPU)**
resize: H, W → multiples of 28 px
(aspect ratio kept, pixel budget
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[P,1176]

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Conv3D 3→1280
kernel = stride
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no bias

[P,1280]

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width 1280 · 16 heads × 80
2D RoPE on q, k (row, col)
window attention: 8×8 patches
full attention: blocks 7, 15, 23, 31
SwiGLU MLP 1280→3420→1280
bidirectional (no causal mask)

[P,1280]

**Patch merger (projector)**
2×2 neighbouring patches
→ 1 token
RMSNorm, concat 4×1280 = 5120
Linear 5120→5120 · GELU
Linear 5120→3584

[N, 3584]

## TEXT PATH

**Prompt**
question
+ image

→

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<|im_start|>system … <|im_end|>
<|im_start|>user
<|vision_start|><|image_pad|><|vision_end|>
What is in the image?<|im_end|>
<|im_start|>assistant
<|image_pad|> is repeated N times

text

**Tokenizer**
byte-level BPE
151,643 tokens
+ 22 special
= 151,665 ids

[L] ids

**Token embedding**
lookup table V × d
= 152,064 × 3584
rows ≥ 151,665: unused padding
(text tokens → vectors)

[L,3584]

**Insert image features**
the N rows at <|image_pad|>
positions are replaced
by the N merger outputs

[L,3584]

## LANGUAGE MODEL

**3D position ids (M-RoPE)**
text: t = h = w = position
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from token ids + image grid
→ cos / sin used in every layer

[3, L]

**Decoder layer ×28**
pre-RMSNorm + residual
self-attention (GQA, causal)
28 query heads, 4 KV heads
head dim 128, M-RoPE on q, k
SwiGLU MLP 3584→18944→3584
KV cache for decoding

[L, 3584]

**Final RMSNorm**

[1, 3584] last pos.

**LM head**
Linear 3584 → 152,064
no bias
own weights (not tied)

[1, 152,064] logits

**Next token**
softmax → greedy / sampling
only the last position
is needed

*decode loop: the new token is embedded and fed back,*
*one token per step; earlier keys / values come from the KV cache*

### Sizes of this model (from config.json)

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Vision encoder (same shape in 3B and 7B)
  width 1280, 32 blocks, 16 heads × 80
  MLP 1280→3420→1280, window 112 px
Language model
  width d = 3584, 28 layers
  28 q-heads / 4 KV-heads × 128
  MLP 3584→18944→3584
  vocab rows 152,064, tied head: no
  RoPE θ = 1,000,000, RMSNorm ε = 1e-06
```

### Where the parameters are



- ViT (patch embed + 32 blocks): 632.0 M (7.6 %)
- patch merger: 44.6 M (0.5 %)
- token embedding: 545.0 M (6.6 %)
- 28 decoder layers + final norm: 6.53 B (78.7 %)
- LM head: 545.0 M (6.6 %)

bf16 weights: 16.6 GB · KV cache: 56 KiB per token · decoder ≈ 14.1 GFLOPs per generated token (linear layers)

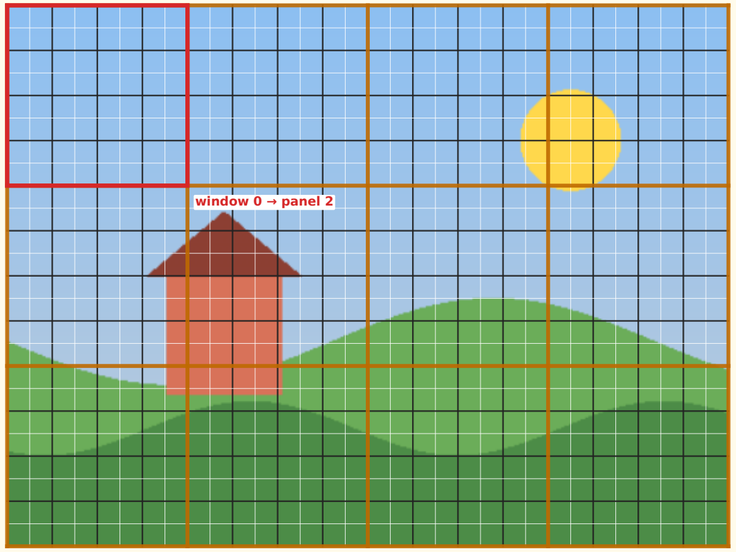
Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# From pixels to visual tokens

What the image processor and the ViT do with a picture (identical in 3B and 7B). Example: a 450×340 photo.

## 1 · Resize, cut into patches, group, window

450×340 → 448×336 px (nearest multiples of 28)




**thin: 14×14 px patch**

32 × 24 = 768 patches (P)

**medium: 28×28 px = 2×2 patches**

= 1 LLM token: 16 × 12 = 192 (N)

**thick: 112×112 px = 8×8 patches**

= 1 attention window: 4 × 3 = 12

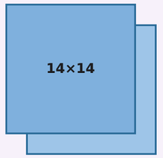
*Windows at the right / bottom edge can be smaller.*

## 2 · Sequence order inside one window

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 4 | 5 | 8 | 9 | 12 | 13 |
| 2 | 3 | 6 | 7 | 10 | 11 | 14 | 15 |
| 16 | 17 | 20 | 21 | 24 | 25 | 28 | 29 |
| 18 | 19 | 22 | 23 | 26 | 27 | 30 | 31 |
| 32 | 33 | 36 | 37 | 40 | 41 | 44 | 45 |
| 34 | 35 | 38 | 39 | 42 | 43 | 46 | 47 |
| 48 | 49 | 52 | 53 | 56 | 57 | 60 | 61 |
| 50 | 51 | 54 | 55 | 58 | 59 | 62 | 63 |

- numbers = position in the ViT sequence (window 0)
- the 4 patches of a 2×2 unit (green outline) are consecutive — the merger later joins rows 4k … 4k+3
- units go row by row inside a window; windows go row by row over the image (window 1 starts at 64)
- window attention: a patch attends to the ≤ 64 patches of its own window (28 of 32 blocks)
- full attention (blocks 7, 15, 23, 31): to all 768
- after the merger, the 192 tokens are put back in plain row-major order (12 rows × 16 columns)

## 3 · One patch → one vector → one token




14 × 14 px, RGB, the image copied into 2 frames (videos: 2 real frames)

**3 · 2 · 14 · 14 = 1176 numbers**

↓

**Patch embedding: Linear 1176 → 1280**

↓

**patch vector, 1280 numbers**

*… 32 vision blocks (shape stays [P, 1280]) …*

**patch 1** | **patch 2** | **patch 3** | **patch 4**

**concat → 5120 numbers (one 2×2 unit)**

↓

**Merger MLP: 5120 → 5120 → d**

↓

**1 visual token, d numbers**

**d = 2048 (3B) · d = 3584 (7B)**

It replaces one <|image_pad|> embedding, so the LLM reads it like any other token.

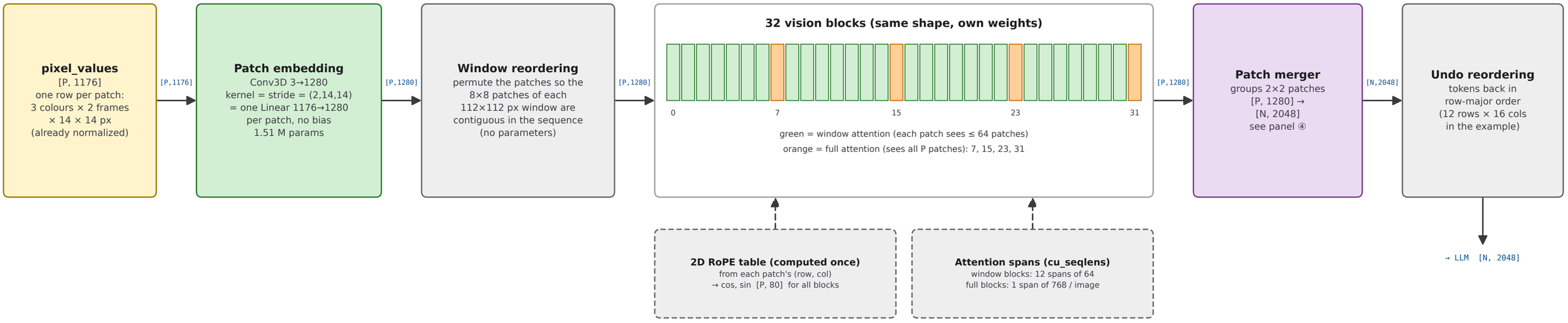
One token covers 28 × 28 px of the image.

Visual tokens N = (H / 28) × (W / 28) e.g. (336 / 28) × (448 / 28) = 12 × 16 = 192 Patches P = 4 N = 768
Default pixel budget: min_pixels 3,136 … max_pixels 12,845,056 → 4 … 16,384 tokens per image. This harness: 256 … 2,048 tokens (jetson/models/qwen2_5_vl/model.sh).

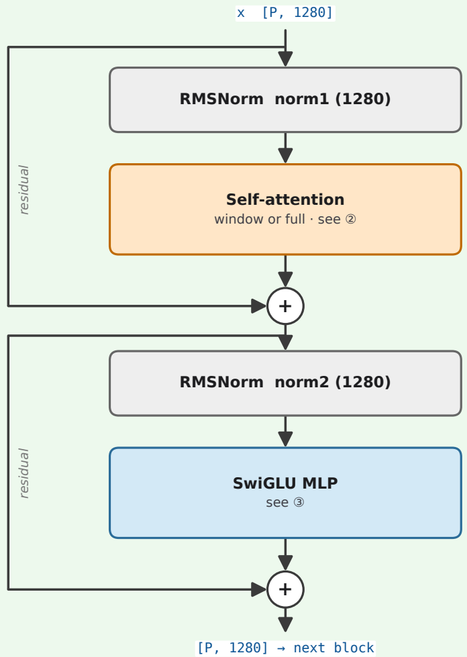
Image processor: Qwen2VLImageProcessor (smart_resize, patch flattening). Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# Qwen2.5-VL-3B-Instruct — vision encoder in detail

ViT 632.0 M + patch merger 36.7 M.  Identical layout in 3B and 7B except the merger's last Linear (5120→2048).  Example numbers: P = 768 patches, N = 192 tokens.

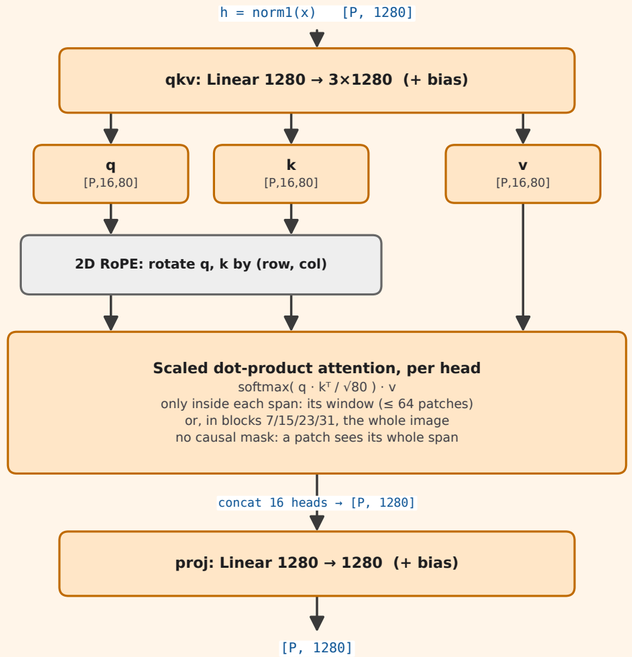



## ① Vision block (×32)



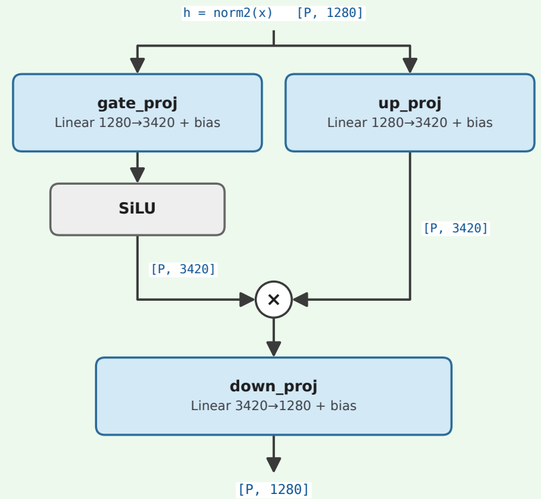

19.70 M params / block (×32 = 630.5 M)

## ② Vision self-attention




6.56 M params / block

## ③ Vision MLP (SwiGLU)

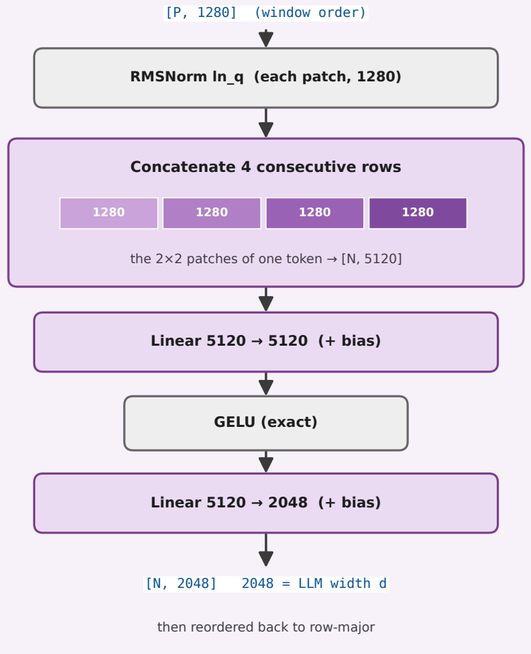



MLP(h) = down( SiLU(gate(h)) ⊙ up(h) )

⊙ = element-wise product: the SiLU branch decides how much of each of the 3420 features passes

13.14 M params / block

## ④ Patch merger



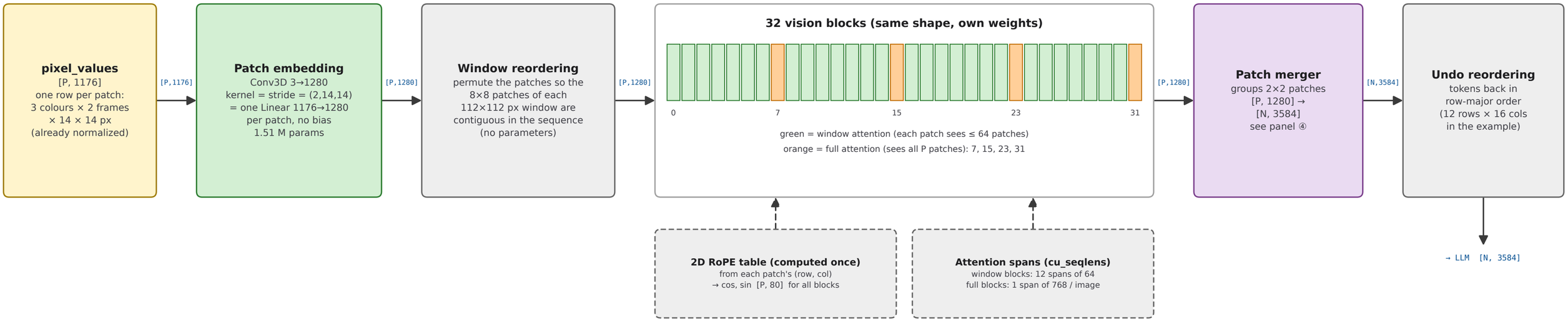

then reordered back to row-major

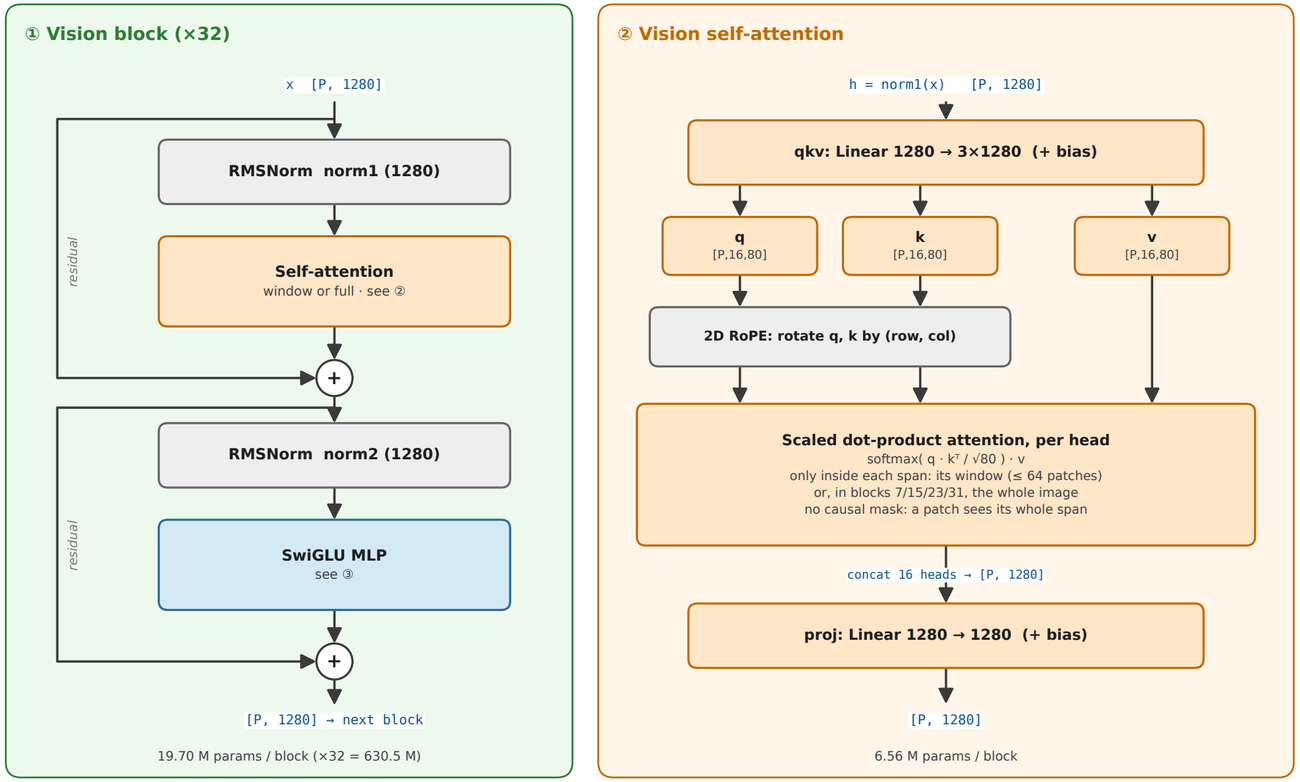
36.71 M params

Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# Qwen2.5-VL-7B-Instruct — vision encoder in detail

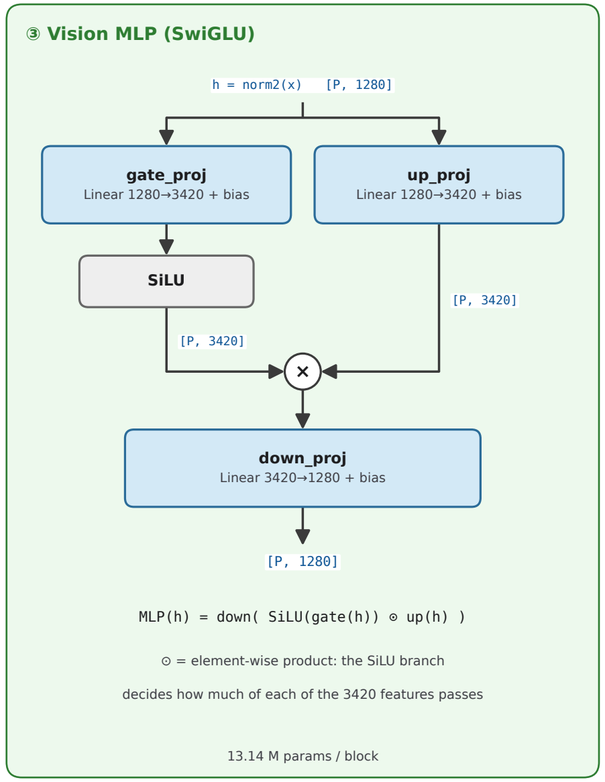
ViT 632.0 M + patch merger 44.6 M.  Identical layout in 3B and 7B except the merger's last Linear (5120→3584).  Example numbers: P = 768 patches, N = 192 tokens.

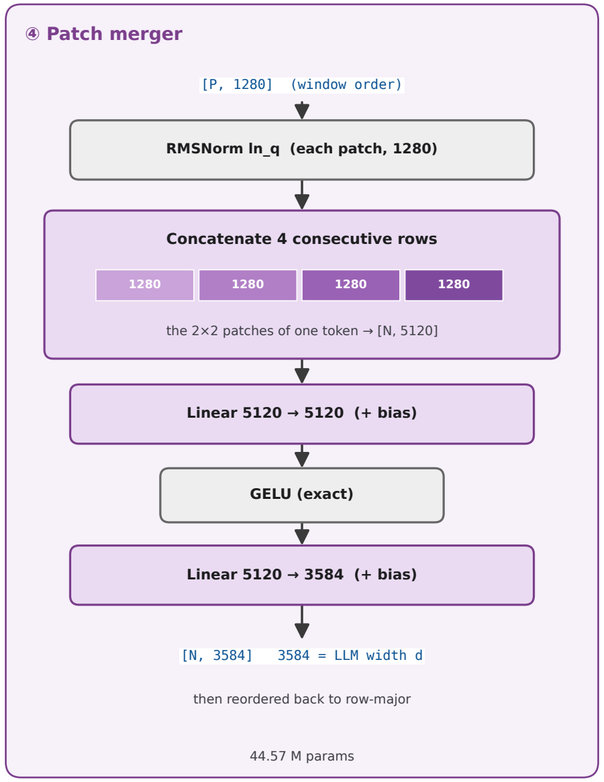








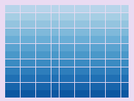




Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# Where is each token?  Rotary position encodings

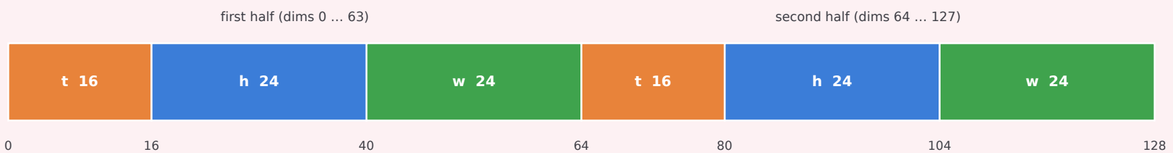
LLM: 3D multimodal RoPE (M-RoPE, head dim 128).  ViT: 2D RoPE (head dim 80).  Same in 3B and 7B.  No learned position embeddings anywhere.

## A · Position ids of the example prompt  (L = 219 tokens; image = 12 × 16 tokens) — values from Qwen2_5_VLModel.get_rope_index

| tokens | <\|im_start\|>system<br>You are a helpful<br>assistant.<\|im_end\|><br><\|im_start\|>user | <\|vision_<br>start\|> |  <br>192 × <\|image_pad\|><br>12 rows (h) × 16 columns (w)<br>row-major, as the merger outputs them | <\|vision_<br>end\|> | What is in the<br>image?<\|im_end\|><br><\|im_start\|><br>assistant | 1st new<br>token |
|---|---|---|---|---|---|---|
| **sequence index** | 0 … 13 | 14 | 15 … 206 | 207 | 208 … 218 | 219 |
| **t  (time)** | **0 … 13** | **14** | **15  (all 192)** | **31** | **32 … 42** | **43** |
| **h  (height)** | **0 … 13** | **14** | **15 + row = 15 … 26** | **31** | **32 … 42** | **43** |
| **w  (width)** | **0 … 13** | **14** | **15 + col = 15 … 30** | **31** | **32 … 42** | **43** |

- Text tokens get t = h = w = their running position, so for pure text M-RoPE is exactly the 1D RoPE of Qwen2.5.
- All image tokens share t (one frame) and take h, w from their row / column in the 12 × 16 grid, offset by the start position 15.
- Text after the image resumes at max + 1 = 31: the 192 image tokens advance the position by only 16 (max(rows, cols)), not by 192.
- Decoding: every new token gets t = h = w = sequence index − 176 (the stored rope delta = 43 − 219), e.g. index 219 → 43, 220 → 44.
- Videos: t = frame-group index × seconds per group × 2 (tokens_per_second), so t follows real time.

## B · LLM: which of the 128 dims of a head see t, h or w



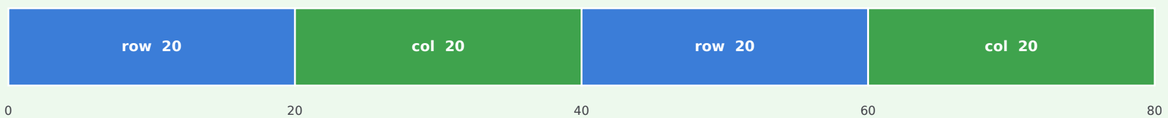

RoPE rotates dim i together with dim i + 64, at frequency θ^(−2i/128), θ = 10⁶.

Of the 64 rotation pairs, the 16 fastest use t, the next 24 use h,

the slowest 24 use w   (rope_scaling.mrope_section = [16, 24, 24]).

Applied to q and k in every layer; v is not rotated.

## C · ViT: 2D RoPE inside the 80 dims of a vision head




Dim j rotates with dim j + 40. Row and column each use the same 20

frequencies 10000^(−2k/40), k = 0 … 19  (θ = 10⁴).

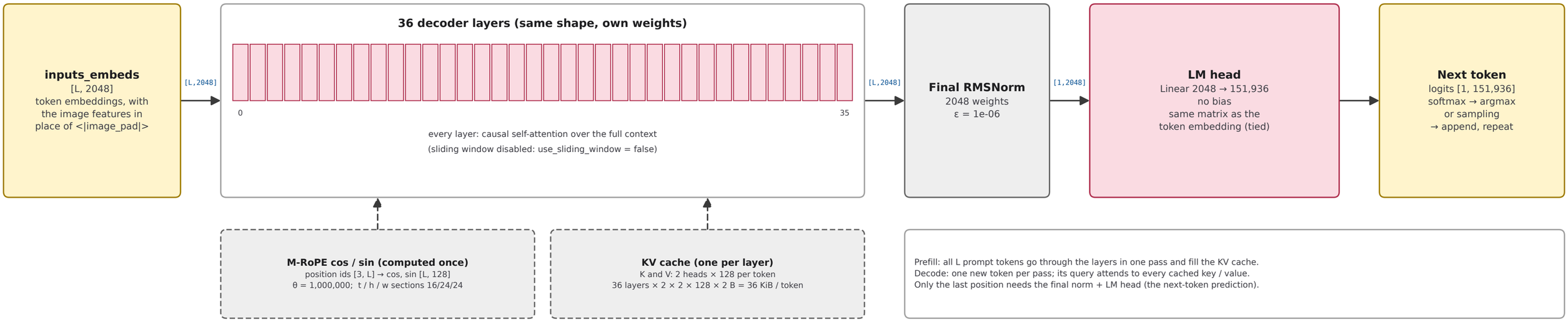
Positions = the patch's (row, col) in the 14-px grid of its own image.

cos / sin are computed once and reused by all 32 blocks.

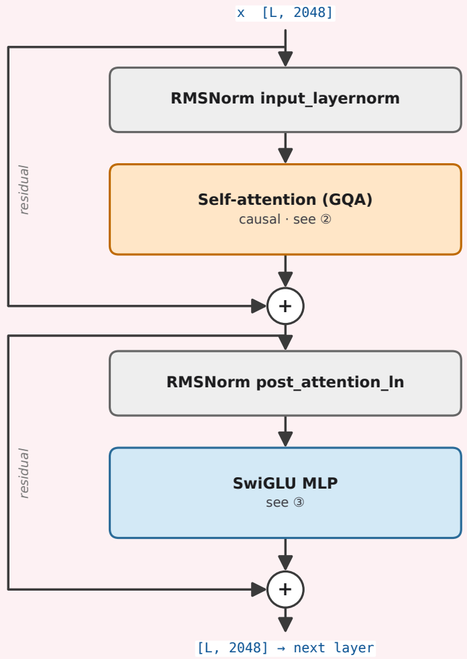
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# Qwen2.5-VL-3B-Instruct — language model in detail

Qwen2.5 decoder: 3.09 B parameters, width 2048, 36 layers, 16 query heads / 2 KV heads × 128, vocab rows 151,936, embedding tied with the LM head

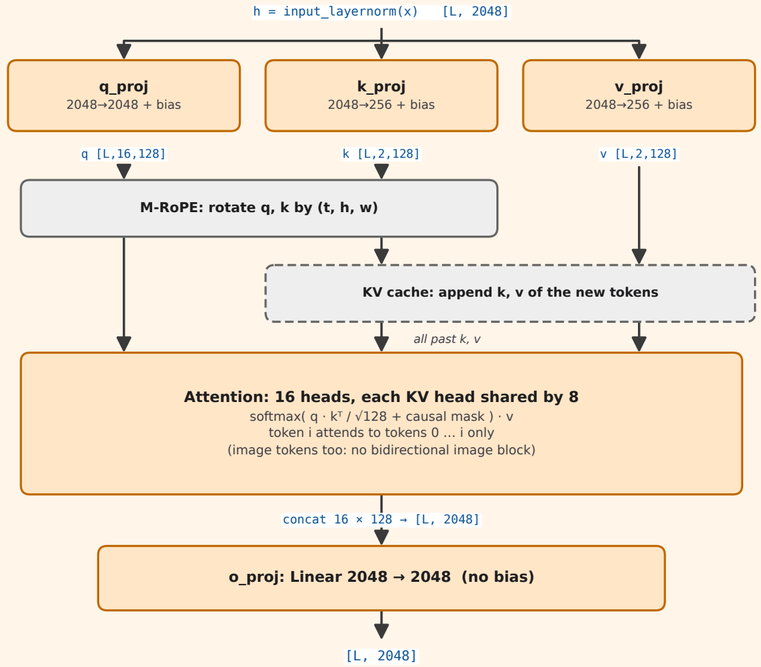



## ① Decoder layer (×36)



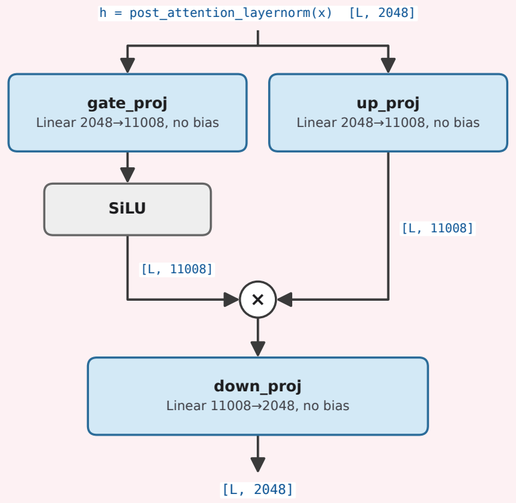

77.08 M params / layer (×36 = 2.77 B)

## ② Grouped-query self-attention




9.44 M params / layer

## ③ MLP (SwiGLU)

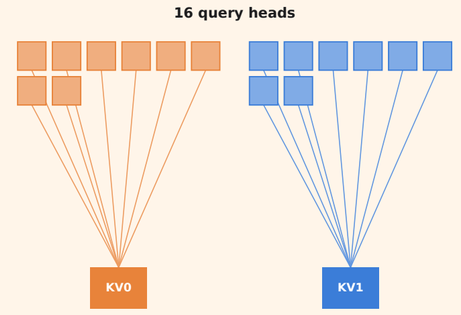



MLP(h) = down( SiLU(gate(h)) ⊙ up(h) )

⊙ = element-wise product: the SiLU branch decides how much of each of the 11008 features passes

67.63 M params / layer

## ④ Who shares what (GQA)




**2 key/value heads**

Each group of 8 query heads reads the same key and value head.

K / V cache per token and layer:

2 heads × 128 × 2 (K, V) × 2 bytes

= 1,024 B → × 36 layers

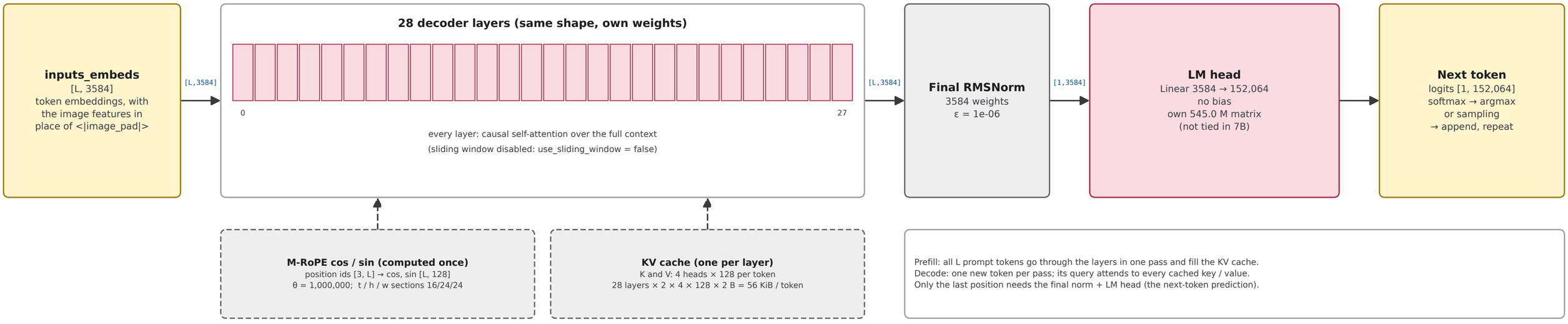
= 36 KiB per token

(8× smaller than with 16 KV heads)

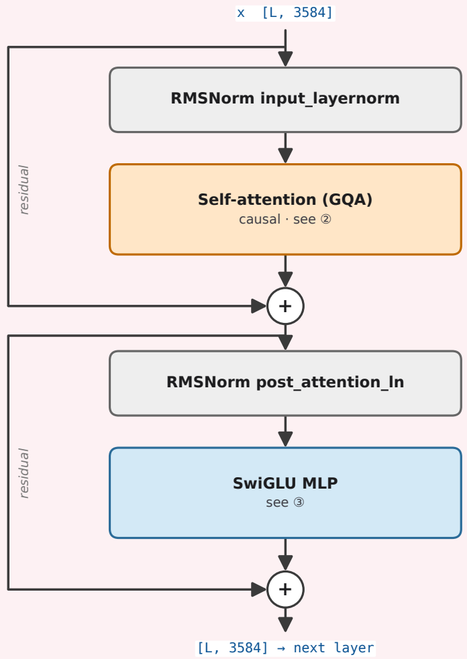
Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# Qwen2.5-VL-7B-Instruct — language model in detail

Qwen2.5 decoder: 7.62 B parameters, width 3584, 28 layers, 28 query heads / 4 KV heads × 128, vocab rows 152,064, separate LM head

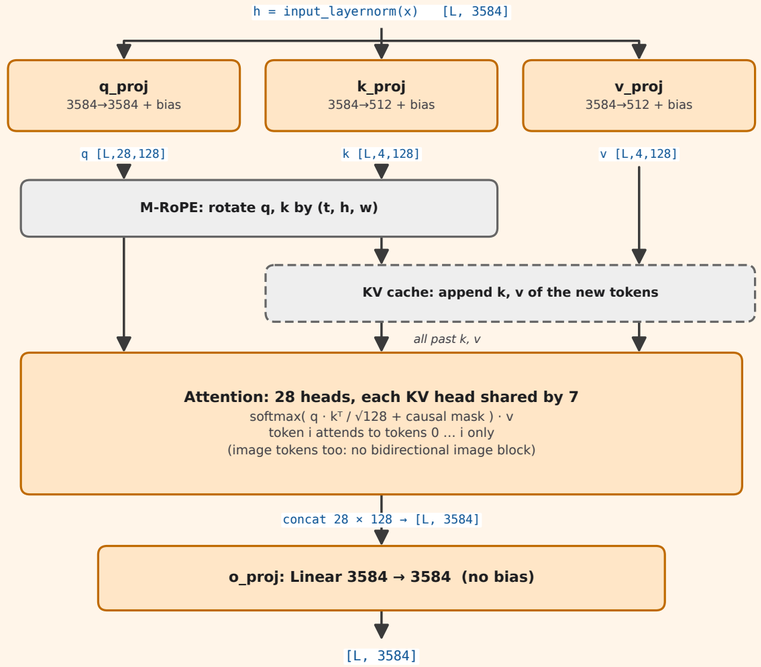



## ① Decoder layer (×28)



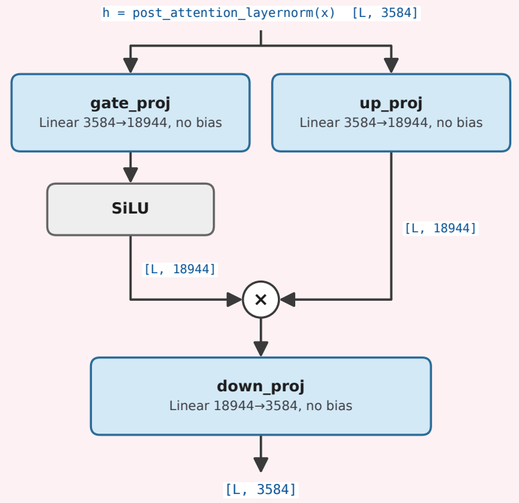

233.06 M params / layer (×28 = 6.53 B)

## ② Grouped-query self-attention




29.36 M params / layer

## ③ MLP (SwiGLU)

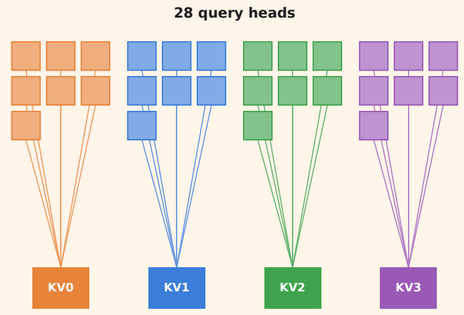



`MLP(h) = down( SiLU(gate(h)) ⊙ up(h) )`

⊙ = element-wise product: the SiLU branch decides how much of each of the 18944 features passes

203.69 M params / layer

## ④ Who shares what (GQA)




Each group of 7 query heads reads the same key and value head.

K / V cache per token and layer:

4 heads × 128 × 2 (K, V) × 2 bytes

= 2,048 B → × 28 layers

= 56 KiB per token

(7× smaller than with 28 KV heads)

Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py

# Qwen2.5-VL-3B vs 7B — what changes

Same recipe; the vision encoder keeps its shape, the language model is wider (7B) or deeper (3B). Highlighted rows differ.

| | 3B | 7B | note |
|---|---|---|---|
| Total parameters | **3.75 B** | **8.29 B** | |
| bf16 weights | **7.5 GB** | **16.6 GB** | 2 bytes per parameter |
| **VISION ENCODER** | | | |
| width · blocks · heads × dim | 1280 · 32 · 16×80 | 1280 · 32 · 16×80 | same shape, separately trained weights |
| MLP (SwiGLU, with bias) | 1280→3420→1280 | 1280→3420→1280 | |
| full-attention blocks | 7, 15, 23, 31 | 7, 15, 23, 31 | the others use 112×112 px windows |
| ViT parameters | 632.0 M | 632.0 M | |
| patch merger | **5120→5120→2048** | **5120→5120→3584** | output = LLM width |
| merger parameters | **36.7 M** | **44.6 M** | |
| **LANGUAGE MODEL** | | | |
| hidden size d | **2048** | **3584** | |
| decoder layers | **36** | **28** | 3B is deeper but much narrower |
| query heads / KV heads | **16 / 2** | **28 / 4** | query heads per KV head: 8 vs 7 |
| head dim | 128 | 128 | |
| MLP intermediate size | **11,008** | **18,944** | 5.38 × d  vs  5.29 × d |
| vocab rows (embedding / LM head) | **151,936** | **152,064** | 151,665 real tokens, the rest padding |
| LM head | **tied to embedding** | **separate (+545.0 M)** | |
| RoPE θ · M-RoPE sections | 1,000,000 · [16, 24, 24] | 1,000,000 · [16, 24, 24] | |
| language-model parameters | **3.09 B** | **7.62 B** | incl. embedding (and LM head) |
| KV cache per token (bf16) | **36 KiB** | **56 KiB** | layers × 2 × KV heads × 128 × 2 B |
| decoder GFLOPs per token | **6.2** | **14.1** | 2 × (layer + LM head weights), w/o attention |

Sources: config.json of the checkpoint · transformers 4.57 modeling_qwen2_5_vl.py · generated by make_diagrams.py